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DEPARTMENT OF ELECTRICAL ENGINEERING



INTERNET OF THINGS (EE - 325) PROJECT REPORT

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Abstract

This project presents the design and simulation of an IoT-based smart door lock system using RFID and keypad-based authentication methods. The system aims to enhance security and convenience by replacing conventional mechanical locks with a digital access control mechanism. Users can gain access either by scanning an RFID card, identified by its unique identification number (UID), or by entering a password through a keypad as a backup authentication method.

An Arduino microcontroller processes the input data received from the RFID reader and keypad, verifies the credentials with stored authorized values, and accordingly controls a servo motor that simulates the door locking mechanism. Visual and audio feedback is provided using LEDs and a buzzer to indicate access status. The system also includes an automatic door locking feature that locks the door after a predefined time interval once access is granted, improving overall security.

In addition, a master password mechanism is implemented to allow the authorized user to update the access password without modifying the program code. The complete system is developed and tested using Tinkercad and Wokwi simulators, which enable both logic-level and hardware-level validation without the need for physical components. The project demonstrates fundamental embedded system concepts and provides a foundation for future IoT-based smart security applications.

Introduction

The increasing adoption of smart homes and IoT technologies has created a demand for intelligent security systems that offer better protection and convenience than traditional mechanical locks. Conventional locks depend on physical keys, which can be lost, duplicated, or misused, thereby posing security risks. Smart locking systems overcome these limitations by using electronic authentication methods such as RFID, keypads, and mobile-based access control. These systems provide improved security, easier access management, and scope for monitoring and automation.

RFID technology enables contactless identification of users using unique card identifiers, making it suitable for access control applications. However, relying solely on RFID can be limiting in cases of card loss or device malfunction. Therefore, a keypad-based authentication method is included as a backup mechanism in this project. The system is built around an Arduino microcontroller, which receives input from the RFID reader and keypad, processes authentication logic, and controls a servo motor to lock or unlock the door. LEDs and a buzzer provide immediate feedback to the user, while an automatic locking feature ensures the door is not left unlocked for extended periods.

The project is implemented using simulation platforms, allowing safe and cost-effective testing of both software logic and hardware interfacing. The proposed system reflects real-world smart lock functionality and can be extended in the future to include features such as mobile app control, cloud-based access logging, and integration with smart home systems.

1. Objectives

- To design a smart door lock system using RFID and keypad authentication
- To simulate the working of a real smart lock using online simulators
- To implement a master password feature for changing user passwords
- To include an automatic door locking mechanism for security
- To demonstrate basic embedded system concepts and access control logic

2. System Overview

The smart lock system consists of an input unit (RFID reader and keypad), a processing unit (Arduino microcontroller), and output units (servo motor, LEDs, and buzzer). The system operates in two modes:

- **Normal Mode:** User authentication and door access
- **Set Password Mode:** Owner authentication and password update

On successful authentication, the door unlocks for a fixed time and then automatically locks. On invalid authentication, a buzzer and red LED indicate access denial.

3. Hardware and Software Requirements

Hardware(Simulated):

- Arduino Uno
- RC522 RFID Reader
- 4×4 Keypad
- Servo Motor
- Red and Green LEDs
- Buzzer
- Connecting Wires

Software/Tools:

- Tinkercad Circuits
- Wokwi Simulator
- Arduino IDE
- MFRC522 RFID Library

4. Block Diagram (Description)

The block diagram consists of:

RFID Reader / Keypad → Arduino Uno → Servo Motor (Door Lock)

→ LEDs (Status)

→ Buzzer (Alarm)

5. Working Principle

1. The user scans an RFID card or enters a password using the keypad.
2. The Arduino reads the UID or password and compares it with stored values.
3. If authentication is successful, the servo motor unlocks the door.
4. After a predefined time delay, the door automatically locks again.
5. If authentication fails, a buzzer sounds and a red LED is turned ON.
6. The master password allows changing the user password using the keypad.

6. Algorithm

1. Initialize RFID reader, keypad, servo, LEDs, and buzzer
2. Wait for RFID scan or keypad input
3. If credentials match → unlock door
4. Else → trigger alarm
5. If master password is entered → enter password change mode
6. Update password and return to normal mode
7. After unlock, wait for fixed time and auto-lock the door

7. Results and Observations

- The RFID reader successfully detects virtual RFID card UIDs in Wokwi.
- The keypad provides an alternative authentication method.
- The door unlocks correctly for valid credentials and auto-locks after the set delay.
- Invalid access attempts trigger buzzer and red LED.
- Password update using the master password works correctly.

8. Applications

- Smart home door lock systems
- Office access control
- Hostel/PG room security
- Locker systems
- Restricted area access control

9. Advantages

- Improved security compared to traditional locks
- Contactless access using RFID
- Backup authentication using keypad
- Automatic locking feature
- Can be extended to IoT systems

10. Limitations

- Simulation-based implementation
- Single RFID UID supported in current version
- No cloud logging or mobile app control
- No real hardware deployment

11. Future Scope

- Mobile app control using Bluetooth/Wi-Fi
- Cloud-based access logs
- Support for multiple RFID cards
- Biometric authentication
- Integration with full smart home systems

12. Results and Observations

- The RFID reader successfully detects virtual RFID card UUIDs in Wokwi.
- The keypad provides an alternative authentication method.
- The door unlocks correctly for valid credentials and auto-locks after the set delay.
- Invalid access attempts trigger buzzer and red LED.
- Password update using the master password works correctly

Conclusion

The IoT-based smart lock system was successfully designed and simulated using RFID and keypad authentication. The project demonstrates secure access control, password management, and automatic door locking using embedded system principles. The simulation closely represents real-world smart lock systems and can be further extended with IoT features for practical deployment.